Solar Powered Sensor



The Temco Controls Solar-Powered LoRa Environmental Sensor Node is a low-power wireless sensing device based on the RA-08 LoRa module. It collects environmental data such as temperature, humidity, CO2, TVOC, and light level, then transmits the data wirelessly to a T3 controller equipped with an RA-08 LoRa gateway module.

The sensor node is powered by a solar panel and uses a supercapacitor as the energy storage device. To reduce power consumption, the node keeps most sensors powered off during standby, wakes up periodically, enables the required sensors, collects data, sends the LoRa packet, and then returns to low-power mode.

The T3 controller gateway receives the wireless data and makes it available in the Temco system, including T3000 monitoring and trend log recording.

Highlights

- Solar-powered wireless environmental sensor node
- RA-08 LoRa long-range wireless communication
- Supports temperature, humidity, CO2, TVOC, and light sensing
- Supercapacitor-based energy storage
- Low-power firmware design with periodic wake-up and reporting
- Sensor power control to reduce standby current
- Designed to work with a T3 controller RA-08 LoRa gateway
- Sensor data can be displayed and recorded in T3000 trend logs
- Suitable for remote or hard-to-wire monitoring locations

Specifications

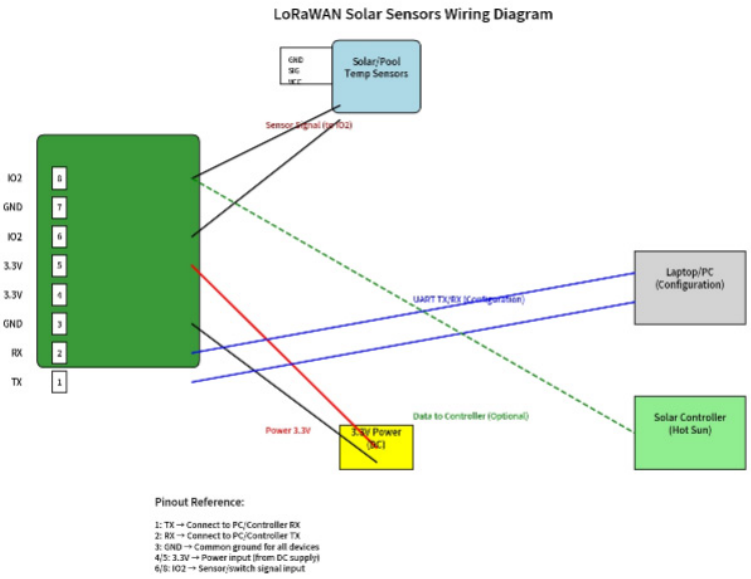
	Relative Humidity	Temperature
Sensor	Capacitance polymer	10K internal
Range	0~100% non condensing	-30~70°C(-22~158°F)
Output	4-20mA, 0-5V or 0-10V, RS 485	4-20mA, 0-5V or 0-10V, RS 485
Accuracy	±2 %@(0-90%)	±0.2C@5-60Degc
Hysteresis	< ±1% RH	
Response time	< 10s (25°C, in slow air)	
Drift	< ±0.5% RH / year	

Parameter	Conditions	Min.	Typ.	Max.	Unit
CO2 output range		380		32’000	ppm
CO2 measurement accuracy1 continuous measurement mode	400 – 5’000 ppm, SHT4x temperature and humidity compensation		± (100 ppm + 10% of reading)		ppm
CO2 measurement accuracy1 single shot measurement mode	400 – 5’000 ppm, SHT4x temperature and humidity compensation, 10 s sampling interval		± (100 ppm + 10% of reading)		ppm
Digital resolution			1		ppm
Response time2	τ63%, step change 2’000 – 400 ppm		20		s

Total volatile organic compounds (TVOC) and why this quantity is related to indoor air quality (IAQ) and the so called IAQ levels. Since Sensirion’s SGP gas sensor is responsive to a broad range of volatile organic compounds (VOC) and other gases relevant for indoor air quality, the present gas sensing technology is well suited for monitoring TVOC concentrations and for translating those into IAQ levels. In order to meet Sensirion’s high quality standards, each SGP sensor is production calibrated.

TVOC (= Total Volatile Organic Compounds) corresponds to the sum of volatile organic co pounds (VOC1). The sum of VOC concentrations, or simply TVOC2, is used as an indication for VOC contamintation. VOC contamination is an established concept in regulatory and scientific literature. Note that the specfic TVOC composition varies between different ambient indoor environments and indoor air is always composed of different volatile organic substances3. Therefore, it is helpful to consider TVOC concentrations as statistical reference values which help to indicate indoor air quality

Wiring Diagram



T3 Controller Gateway Integration

The RA-08 gateway installed in the T3 controller receives wireless packets from the solar sensor nodes. The received values can be mapped into the T3 controller system, exposed to the central BACnet system, and viewed or logged through T3000. Receives LoRa sensor data from multiple remote nodes

- Displays live sensor values in T3000
- Records sensor data in T3000 trend logs
- Allows received values to be used in controller logic, depending on firmware support
- Provides a bridge between wireless sensor nodes and the central BACnet building automation system

Parameter	Min	Typ	Max	Unit	Notes
System Supply Voltage	3.0	3.3	3.6	V	Internal regulated supply
LoRa TX Power	-	20	22	dBm	Depends on module configuration
Operating Temperature	-40	-	+85	deg C	Electronics rating; sensor limits may vary
Sleep Current	TBD	TBD	TBD	uA	To be confirmed by testing
Active Sensor Current	TBD	TBD	TBD	mA	Depends on enabled sensors
LoRa TX Current	TBD	TBD	TBD	mA	Depends on TX power and radio configuration

Internal Interface Signals

Signal	Function
I2C_SCL	I2C clock for environmental sensors
I2C_SDA	I2C data for environmental sensors
CO2_EN	CO2 sensor power enable
SHT40_EN	Temperature/humidity sensor power enable
TVOC_EN	TVOC sensor power enable
LIGHT_EN	Light sensor power enable, if used
LED	Status indicator
SOLAR_IN	Solar panel input
CAP	Supercapacitor connection

Typical Operation Sequence

1.The solar panel charges the supercapacitor during available light.
2.The node remains in low-power mode most of the time.
3.At the configured interval, the node wakes up.
4.Sensor power is enabled.
5.Temperature, humidity, CO2, TVOC, and light data are collected.
6.The node sends the data packet through LoRa.
7.The T3 RA-08 gateway receives the packet.
8.The T3 controller updates the corresponding values.

Installation and Commissioning Notes

- Install the node where the solar panel can receive sufficient light for the required reporting interval.
- Avoid mounting locations with long periods of shade unless a larger energy storage device is used.
- Verify that the T3 controller has the RA-08 gateway module installed and configured.
- Confirm that the node is joined or assigned according to the firmware pairing procedure.
- Check live values in T3000 before enabling trend log recording.
- For long-term testing, record solar input, supercapacitor voltage, reporting interval, and packet reception quality.

Notes

- Sensor configuration, reporting interval, and power behavior may vary by firmware version.
- Solar performance depends strongly on installation location and available light.
- Final current consumption and operating duration should be confirmed with production hardware testing.
- This product is intended to work as part of the Temco T3 controller, T3000, and BACnet building automation ecosystem.